

Physical Chemistry (II)

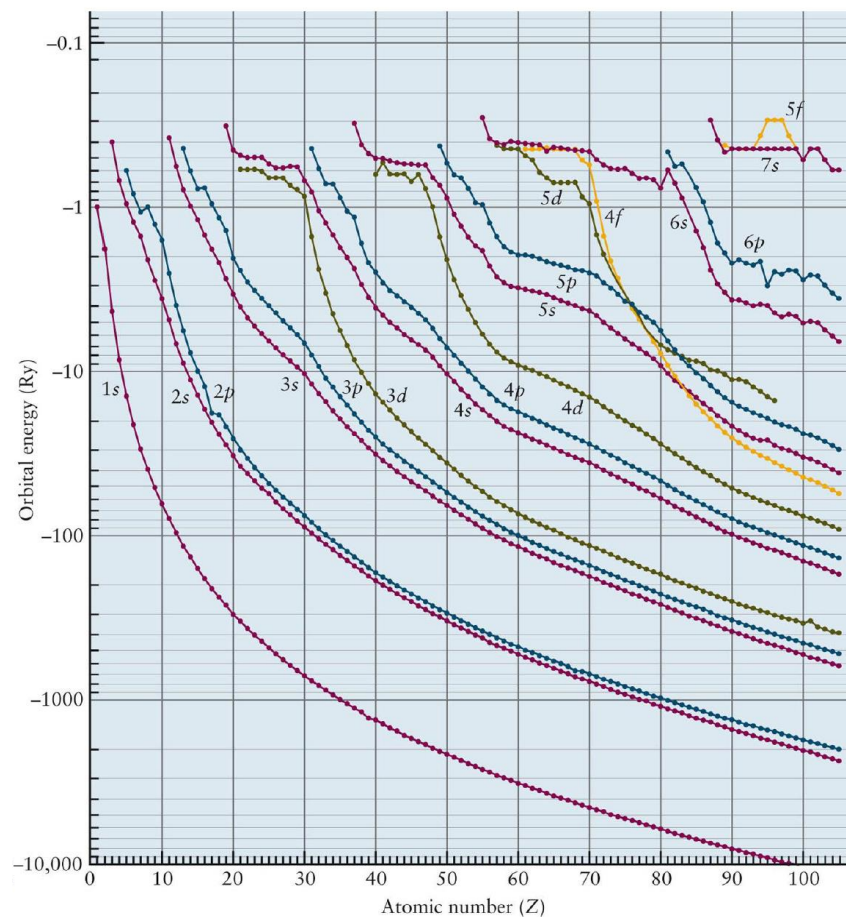
Lecture 20

The Chemical Bond



Last class

Orbital Energies



* e-e repulsion leads to *l*-splitting

Electronic States of Atoms

- Ground-state electron configuration of C: $1s^2 2s^2 2p^2$
→ Same configurations can give “different” states
- Angular momentum is a key to the good description
 - For each electron i , we have $\vec{L}_i$ and $\vec{S}_i$ (and $\vec{J}_i = \vec{L}_i + \vec{S}_i$)
- Two possible ways to obtain quantum numbers
 - Russell-Saunders coupling: $\vec{L}_{\text{total}}$ and $\vec{S}_{\text{total}} \rightarrow \vec{J}_{\text{total}}$
 - j-j coupling: $\vec{J}_i$ for each electron $\rightarrow \vec{J}_{\text{total}}$

Atomic Term Symbol

$$^{2S+1}[L]_J$$

- Symbols for $[L]$

$L =$	0	1	2	3	4	...
	S	P	D	F	G	

- Closed (sub)shells do not contribute
→ only **open subshells** need to be counted

Hund's Rules

Empirical rules to find the ground-state term symbol

1. The state with the **largest S** is the most stable
2. For states with the same S , the state with the **largest L** is the most stable
3. For states with the same S and L ,
 1. If a subshell is **less than half-filled**, the state with the **smallest J** is the most stable
 2. If a subshell is **more than half-filled**, the state with the **largest J** is the most stable



Last class

Selection Rules

(for multi-electron atoms)

$$\Delta L = 0, \pm 1 \quad (L = 0 \nrightarrow L = 0)$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \quad (J = 0 \nrightarrow J = 0)$$

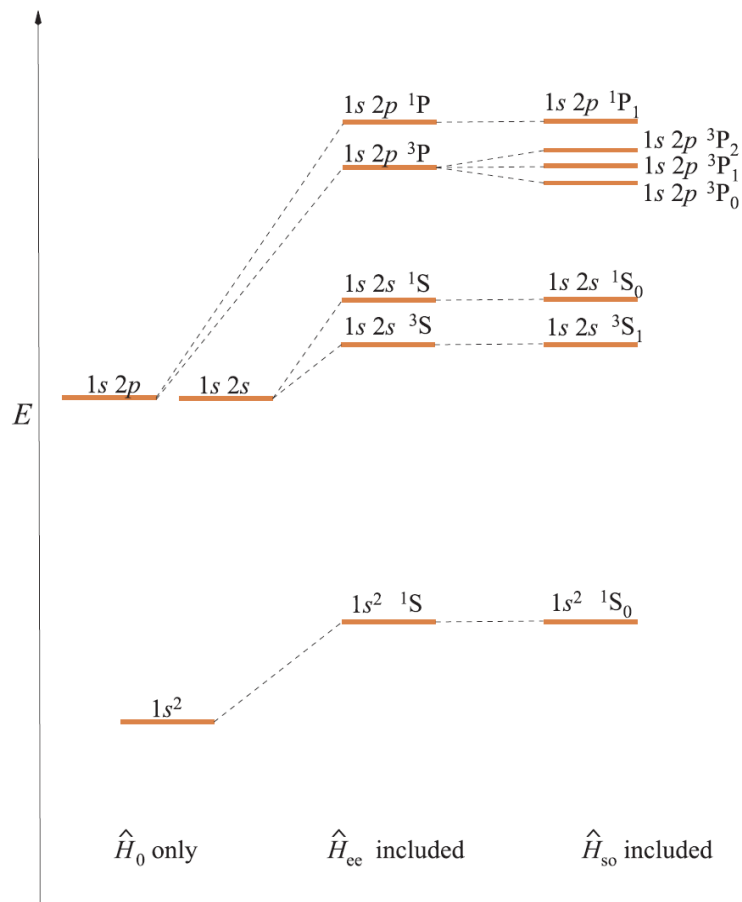


Relativistic Effects

- Origin of the fine structure of H atom
- The spin-orbit coupling is also often called as relativistic effect
- Both effects become more significant as Z increases
- Larger spin-orbit coupling: j - j coupling
- Breakdown of the selection rules

Last class

Energy Splitting of Helium



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Fig. 9.11)

Born-Oppenheimer Approximation

- Nuclei are **much more massive** than electrons
- Electrons adjust instantaneously to any nuclear motion
- For the fixed nuclear geometry $\vec{R}_1, \vec{R}_2, \dots$, solve the Schrödinger equation:

$$\begin{aligned}\hat{H}_{\text{el}}\psi_{\text{el}}(\vec{r}_1, \vec{r}_2, \dots; \vec{R}_1, \vec{R}_2, \dots) \\ = E_{\text{el}}(\vec{R}_1, \vec{R}_2, \dots)\psi_{\text{el}}(\vec{r}_1, \vec{r}_2, \dots; \vec{R}_1, \vec{R}_2, \dots)\end{aligned}$$

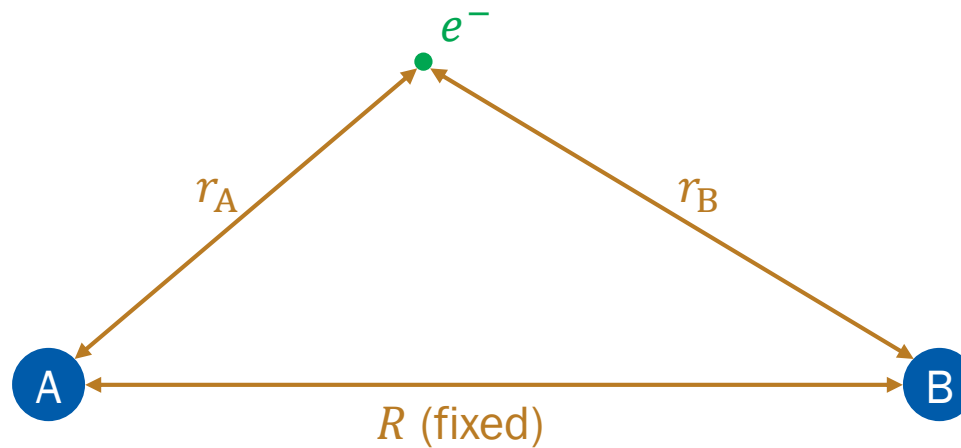
- To solve this, one can apply **any electronic structure calculation method**



RS 10.2

M 10.2

H_2^+ System



$$\hat{H}_{\text{el}} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_A} - \frac{e^2}{4\pi\epsilon_0 r_B} + \frac{e^2}{4\pi\epsilon_0 R}$$

(exactly solvable, but we will use orbitals)

Variational Approach

- Trial function (“molecular orbital”)

$$\phi(\vec{r}) = c_A \psi_{1s,A}(r_A) + c_B \psi_{1s,B}(r_B)$$

where $\psi_{1s,i}(r_i) = \sqrt{\frac{\zeta^3}{\pi}} e^{-\zeta r_i}$ ($\zeta = 1$ below)

- Minimize

$$E_\phi = \frac{\int \phi^* \hat{H} \phi \, d\vec{r}}{\int \phi^* \phi \, d\vec{r}}$$

(Normalization: $|c_A|^2 + |c_B|^2 = 1$)

Numerator

For $\phi = c_A f_A + c_B f_B$ (f_A, f_B : real),

$$\begin{aligned}\int \phi^* \hat{H} \phi \, d\vec{r} &= \int (c_A f_A + c_B f_B) \hat{H} (c_A f_A + c_B f_B) \, d\vec{r} \\ &= c_A^2 H_{AA} + c_A c_B H_{AB} + c_A c_B H_{BA} + c_B^2 H_{BB}\end{aligned}$$

where $H_{ij} = \int f_i \hat{H} f_j \, d\vec{r}$. As for Hermitian $\hat{H}$, $H_{AB} = H_{BA}$,

$$\int \phi^* \hat{H} \phi \, d\vec{r} = c_A^2 H_{AA} + 2c_A c_B H_{AB} + c_B^2 H_{BB}$$

Energy

$$\int \phi^* \hat{H} \phi \, d\vec{r} = c_A^2 H_{AA} + 2c_A c_B H_{AB} + c_B^2 H_{BB}$$

Similarly,

$$\int \phi^* \phi \, d\vec{r} = c_A^2 S_{AA} + 2c_A c_B S_{AB} + c_B^2 S_{BB}$$

where $S_{ij} = \int f_i f_j \, d\vec{r}$.

$$E(c_A, c_B) = \frac{c_A^2 H_{AA} + 2c_A c_B H_{AB} + c_B^2 H_{BB}}{c_A^2 S_{AA} + 2c_A c_B S_{AB} + c_B^2 S_{BB}}$$

Minimization

$$E(c_A, c_B) = \frac{c_A^2 H_{AA} + 2c_A c_B H_{AB} + c_B^2 H_{BB}}{c_A^2 S_{AA} + 2c_A c_B S_{AB} + c_B^2 S_{BB}}$$

Now, minimize $E(c_A, c_B)$ with respect to c_A and c_B :

$$\frac{\partial E}{\partial c_A} = 0 \quad \rightarrow \quad c_A(H_{AA} - ES_{AA}) + c_B(H_{AB} - ES_{AB}) = 0$$

$$\frac{\partial E}{\partial c_B} = 0 \quad \rightarrow \quad c_A(H_{AB} - ES_{AB}) + c_B(H_{BB} - ES_{BB}) = 0$$

Another Use of Determinant

$$c_A(H_{AA} - ES_{AA}) + c_B(H_{AB} - ES_{AB}) = 0$$

$$c_A(H_{AB} - ES_{AB}) + c_B(H_{BB} - ES_{BB}) = 0$$

For nonzero c_A and c_B , E must satisfy

$$\begin{vmatrix} H_{AA} - ES_{AA} & H_{AB} - ES_{AB} \\ H_{AB} - ES_{AB} & H_{BB} - ES_{BB} \end{vmatrix} = 0$$

(secular equation)

Solving the secular equation $\rightarrow$ Optimal E

In General

For a trial function

$$\phi = c_1 f_1 + c_2 f_2 + \cdots + c_N f_N,$$

variational energy is obtained by solving the secular equation:

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} & \cdots & H_{1N} - ES_{1N} \\ H_{21} - ES_{21} & H_{22} - ES_{22} & \cdots & H_{2N} - ES_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ H_{N1} - ES_{N1} & H_{N2} - ES_{N2} & \cdots & H_{NN} - ES_{NN} \end{vmatrix} = 0$$

Two More Conditions

$$\phi(\vec{r}) = c_A \psi_{1s,A}(r_A) + c_B \psi_{1s,B}(r_B)$$

$$\begin{vmatrix} H_{AA} - ES_{AA} & H_{AB} - ES_{AB} \\ H_{AB} - ES_{AB} & H_{BB} - ES_{BB} \end{vmatrix} = 0$$

- Normalization of ψ_{1s} : $S_{AA} = S_{BB} = 1$
- Symmetry of two H atoms: $H_{AA} = H_{BB}$

$$\begin{vmatrix} H_{AA} - E & H_{AB} - ES_{AB} \\ H_{AB} - ES_{AB} & H_{AA} - E \end{vmatrix} = 0$$

RS 10.2

M 10.3

2×2 Determinant

$$\begin{vmatrix} H_{AA} - E & H_{AB} - ES_{AB} \\ H_{AB} - ES_{AB} & H_{AA} - E \end{vmatrix} = 0$$

$$(H_{AA} - E)^2 - (H_{AB} - ES_{AB})^2 = 0$$

Hence,

$$E_{\pm} = \frac{H_{AA} \pm H_{AB}}{1 \pm S_{AB}}$$



Energy and Wavefunction

- Energy

$$E_{\pm} = \frac{H_{AA} \pm H_{AB}}{1 \pm S_{AB}}$$

- Wavefunction

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$

where

$$H_{ij} = \int \psi_{1s,i} \hat{H} \psi_{1s,j} d\vec{r} \text{ and } S_{AB} = \int \psi_{1s,A} \psi_{1s,B} d\vec{r}$$

Integrals

- Overlap integral

$$S_{AB} = \int \psi_{1s,A} \psi_{1s,B} d\vec{r}$$

- Coulomb integral

$$H_{AA} = \int \psi_{1s,A} \hat{H} \psi_{1s,A} d\vec{r}$$

- Exchange integral

$$H_{AB} = \int \psi_{1s,A} \hat{H} \psi_{1s,B} d\vec{r}$$

Integrals Depend on R

- Overlap integral

$$S_{AB}(R) = e^{-R/a_0} \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2} \right)$$

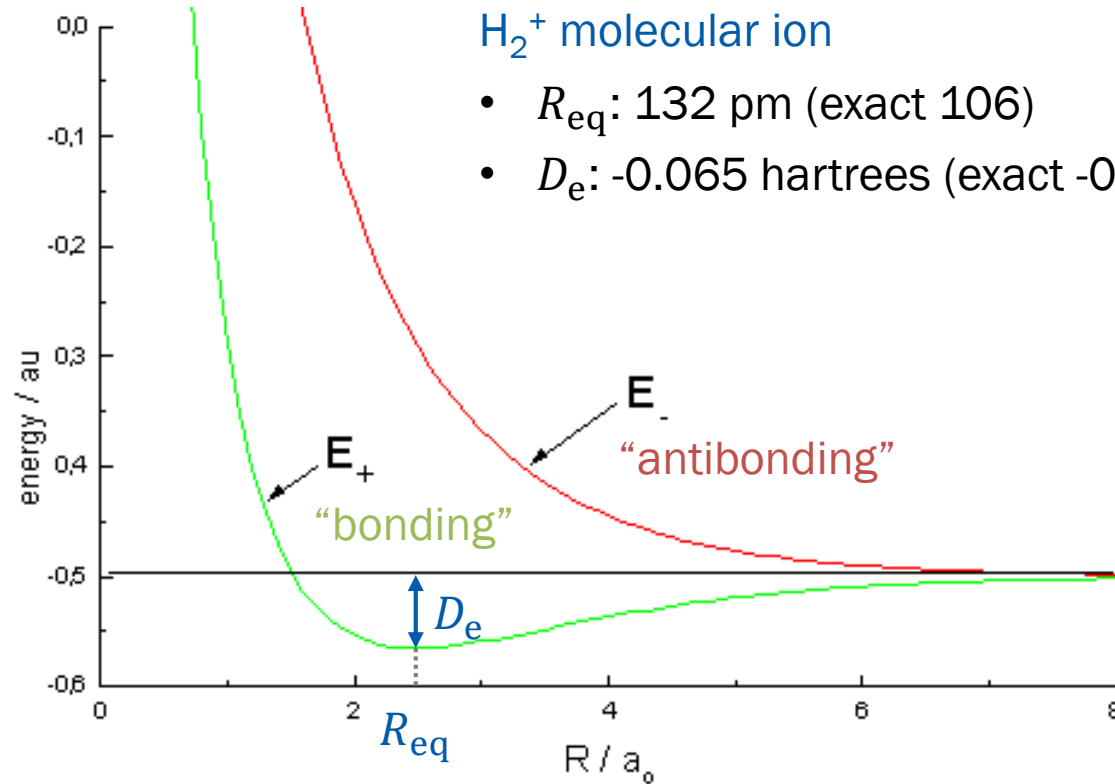
- Coulomb integral

$$H_{AA}(R) = -\frac{1}{2}E_h + e^{-2R/a_0} \left(1 + \frac{a_0}{R} \right) E_h$$

- Exchange integral

$$H_{AB}(R) = \left(\frac{a_0}{R} - \frac{1}{2} \right) S_{AB}(R) E_h - e^{-R/a_0} \left(1 + \frac{R}{a_0} \right) E_h$$

Energy



(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

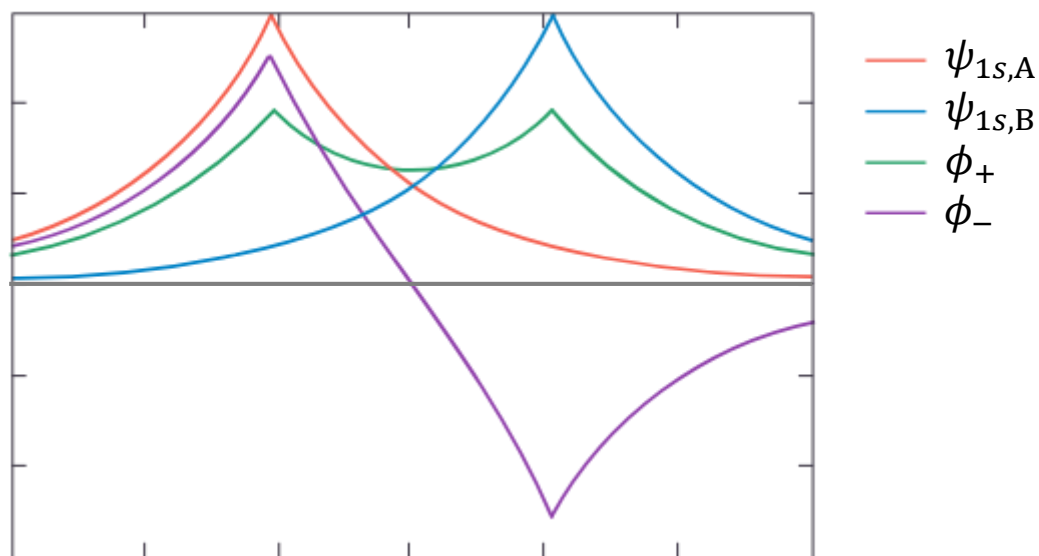
RS 10.2

M 10.4

Wavefunctions

(“Molecular orbitals”)

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$



(Image: <https://chem.libretexts.org/@go/page/1974>)

Multi-Electron Molecules

1. Molecular orbital (MO) theory:

linear combination of atomic orbitals (LCAO)

2. Valence bond (VB) theory

Take H_2 as an example

Ground-State Wavefunction for H₂

- Use H₂⁺ molecular orbitals

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$

- Slater determinant (both electrons reside in ϕ_{+})

$$\psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_{+}(1)|\alpha\rangle_1 & \phi_{+}(1)|\beta\rangle_1 \\ \phi_{+}(2)|\alpha\rangle_2 & \phi_{+}(2)|\beta\rangle_2 \end{vmatrix}$$

$$\psi = \underbrace{\phi_{+}(1)\phi_{+}(2)}_{\text{spatial (MO)}} \times \underbrace{\frac{1}{\sqrt{2}} (|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)}_{\text{spin}}$$

LCAO-MO for H₂

$$\psi_{\text{MO}} = \phi_+(1)\phi_+(2)$$

Substitution leads to

$$\begin{aligned}\psi_{\text{MO}} &= \frac{1}{2(1 + S_{\text{AB}})} [\psi_{1s,\text{A}}(1) + \psi_{1s,\text{B}}(1)] [\psi_{1s,\text{A}}(2) + \psi_{1s,\text{B}}(2)] \\ &= \frac{1}{2(1 + S_{\text{AB}})} [\psi_{1s,\text{A}}(1)\psi_{1s,\text{A}}(2) + \psi_{1s,\text{A}}(1)\psi_{1s,\text{B}}(2) \\ &\quad + \psi_{1s,\text{B}}(1)\psi_{1s,\text{A}}(2) + \psi_{1s,\text{B}}(1)\psi_{1s,\text{B}}(2)]\end{aligned}$$

Energy Calculation

$$\psi_{\text{MO}} = \phi_+(1)\phi_+(2)$$

Since ϕ_+ is the H_2^+ molecular orbital, we can decompose

$$\hat{H} = \underset{\substack{\uparrow \\ \text{independent electrons}}}{\hat{H}_0} + \underset{\substack{\uparrow \\ \text{e-e repulsion}}}{\hat{H}'}$$

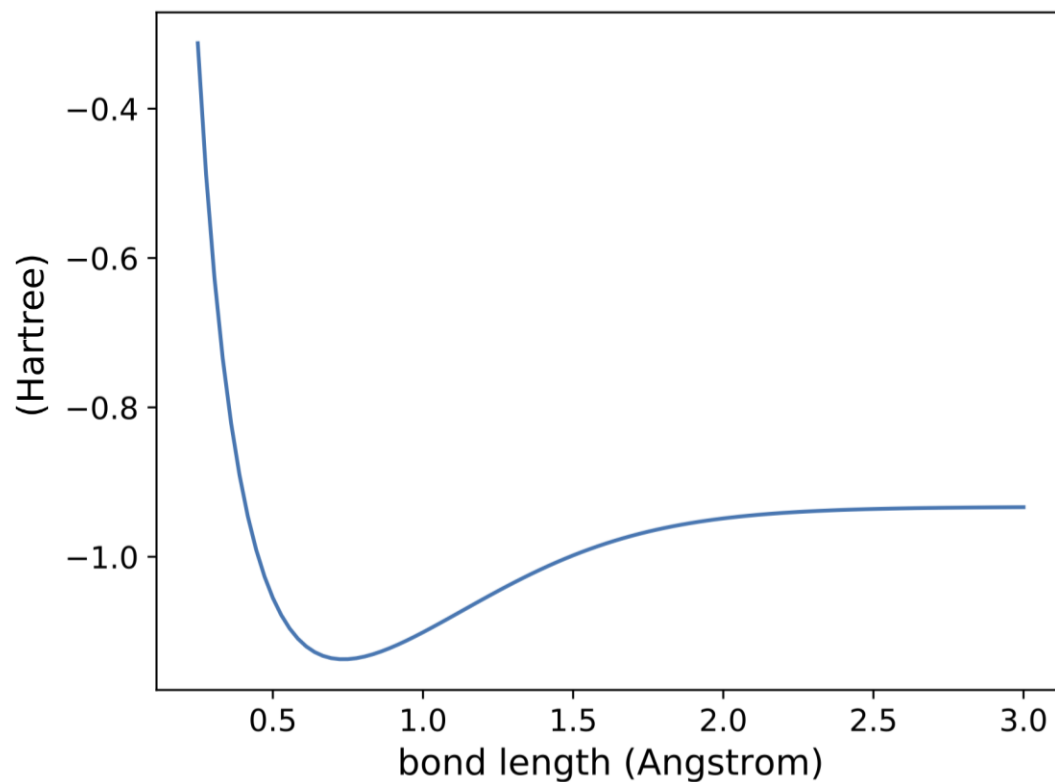
- Perturbation theory

$$\hat{H}_0\psi_{\text{MO}} = E_{\text{MO}}^{(0)}\psi_{\text{MO}}$$

$$E_{\text{MO}}^{(1)} = \int \psi_{\text{MO}} \hat{H}' \psi_{\text{MO}} d\vec{r}_1 d\vec{r}_2$$

RS 10.5
M 10.8

Ground-State Energy of H_2



(Image: <https://quantaggle.com/competitions/gs-pes/>)

Valence Bond Theory

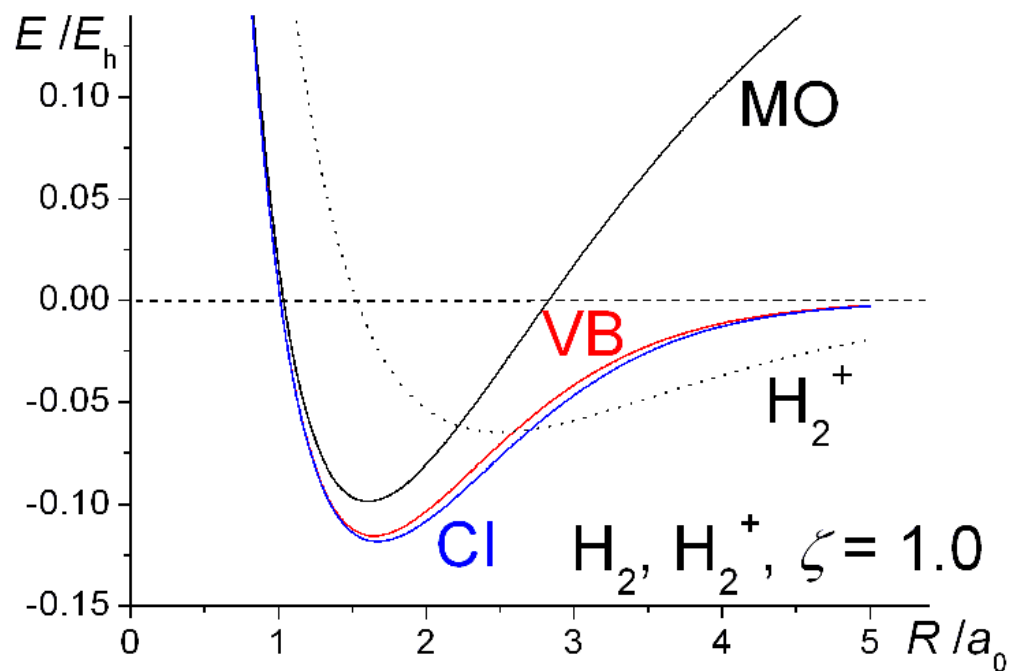
H₂ ground state = two ground-state H atoms

$$\psi_{\text{VB}} = \frac{1}{\sqrt{2(1 + S^2)}} [\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)]$$

We can similarly obtain **energy** by perturbation theory

RS 10.5
M 10.8

Comparison of Models



(Image: Nordholm and Bacskay, *Molecules* 2020)

Origin of Discrepancy

- MO theory

$$\underbrace{\psi_{1s,A}(1)\psi_{1s,A}(2)}_{\text{ionic}} + \underbrace{\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)}_{\text{covalent}} + \underbrace{\psi_{1s,B}(1)\psi_{1s,B}(2)}_{\text{ionic}}$$

- VB theory

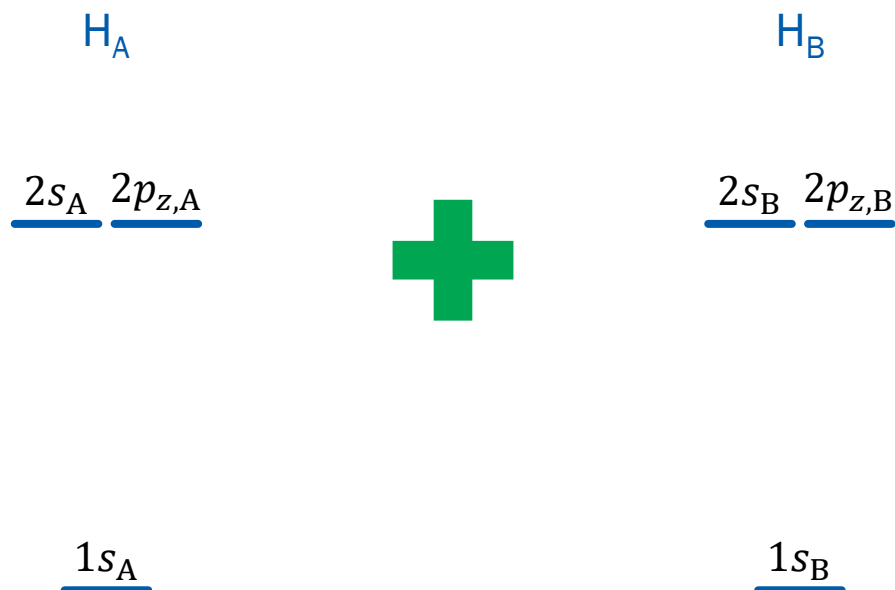
$$\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)$$

- Better model: somewhere between MO and VB

$$\psi = c_1\psi_{\text{VB}} + c_2\psi_{\text{ionic}}$$

Molecular Orbitals

One can combine any number of atomic orbitals
to construct molecular orbitals (LCAO-MO)



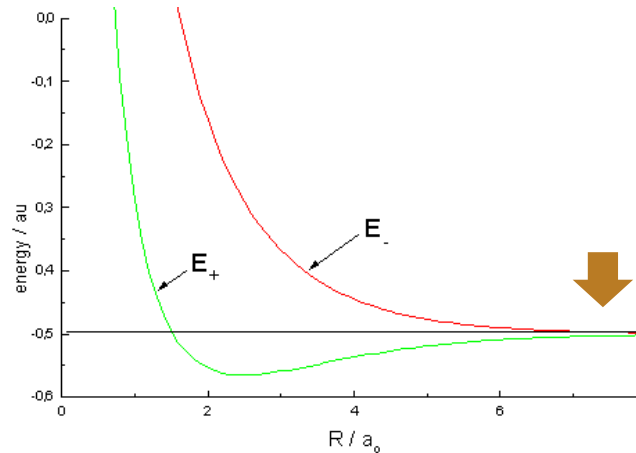
* for fixed nuclear geometry 32



Overlap and Chemical Bond

To make a bond from two atomic orbitals,
we need a significant **overlap** between the two

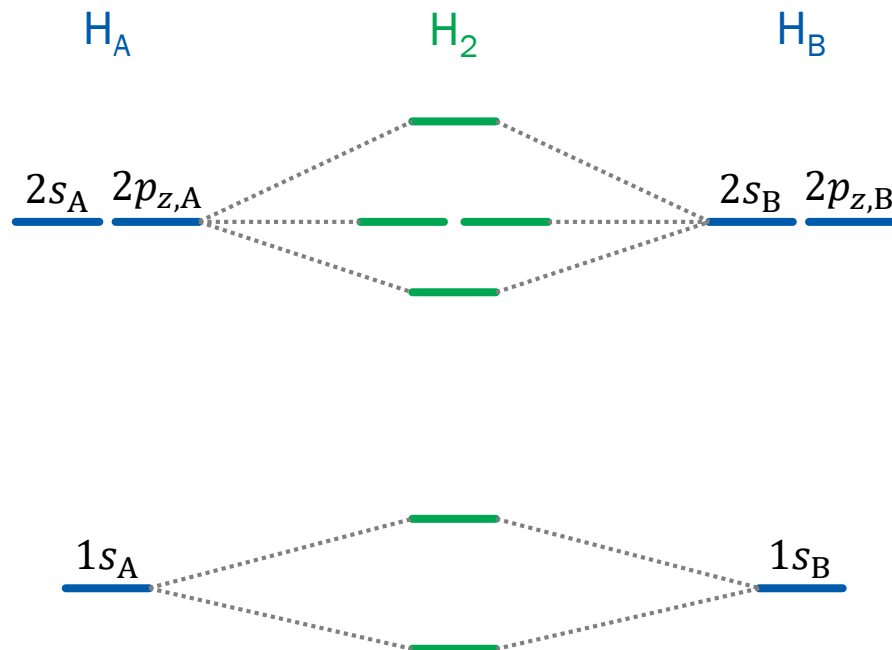
$$S_{AB} = \int \psi_A^* \psi_B d\vec{r} \neq 0$$



(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

Molecular Orbitals

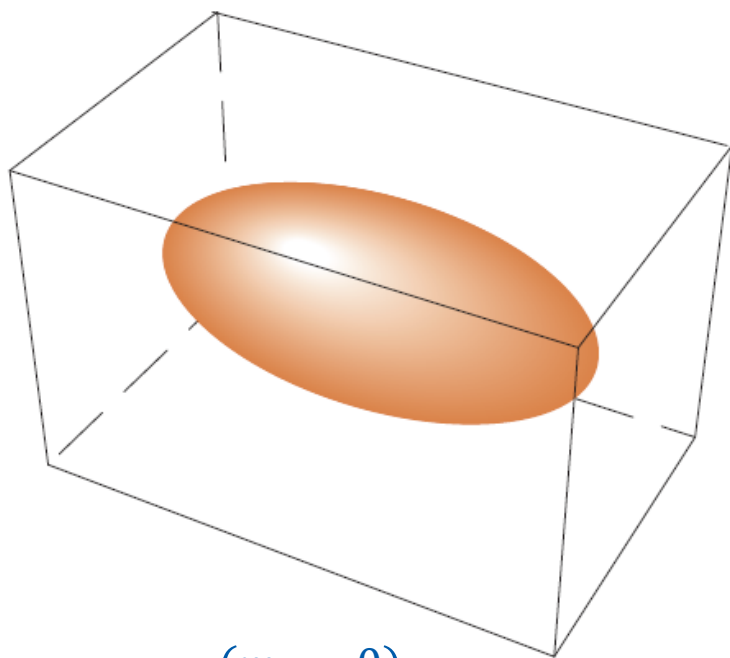
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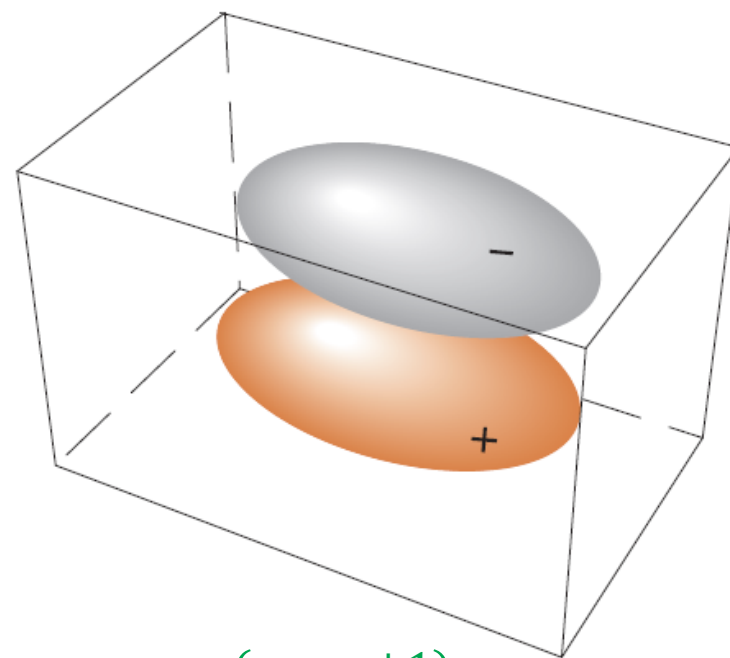
* for fixed nuclear geometry

M 10.2

Symbols for Molecular Orbitals



$\sigma (m_l = 0)$



$\pi (m_l = \pm 1)$

Ingredients of MO Symbols

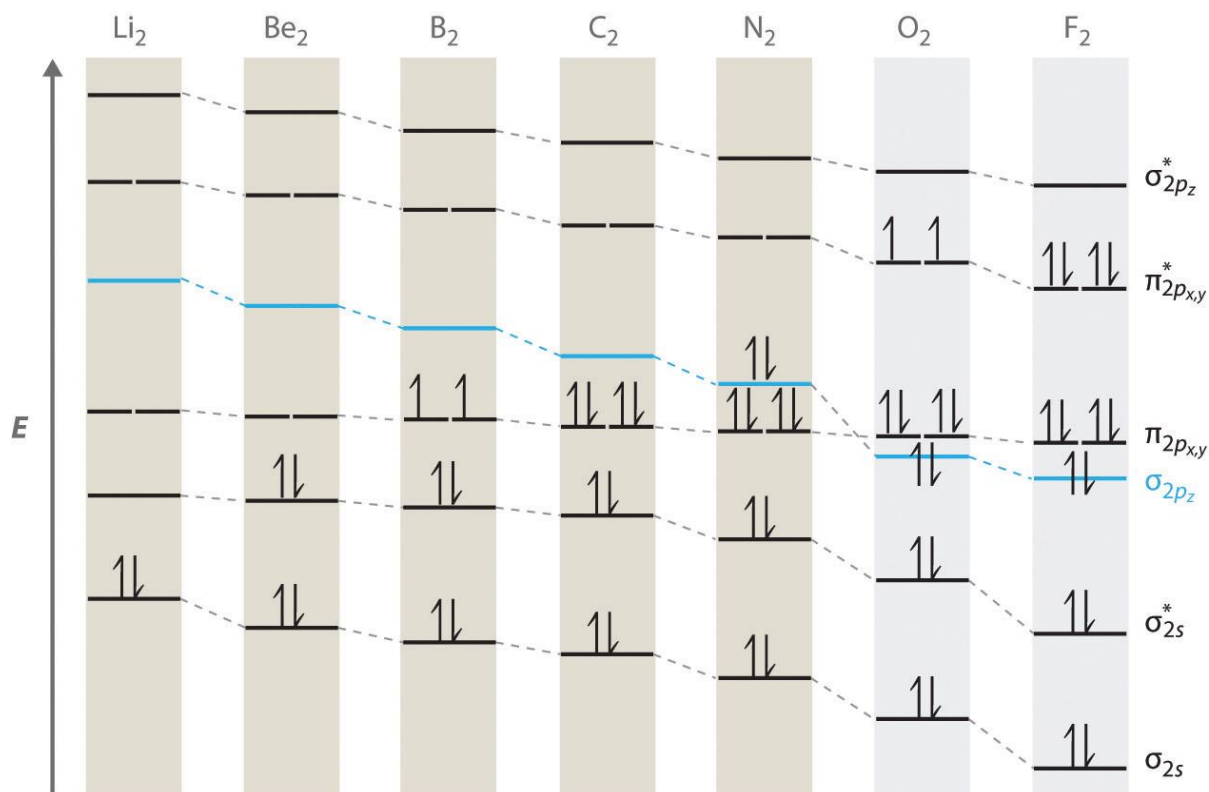
- Change in sign under inversion: g (gerade): no change, u (ungerade): change
- Index: within same type, according to orbital energy
- Atomic orbitals from which MO is constructed
- Bonding, antibonding (*)



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 10.8)

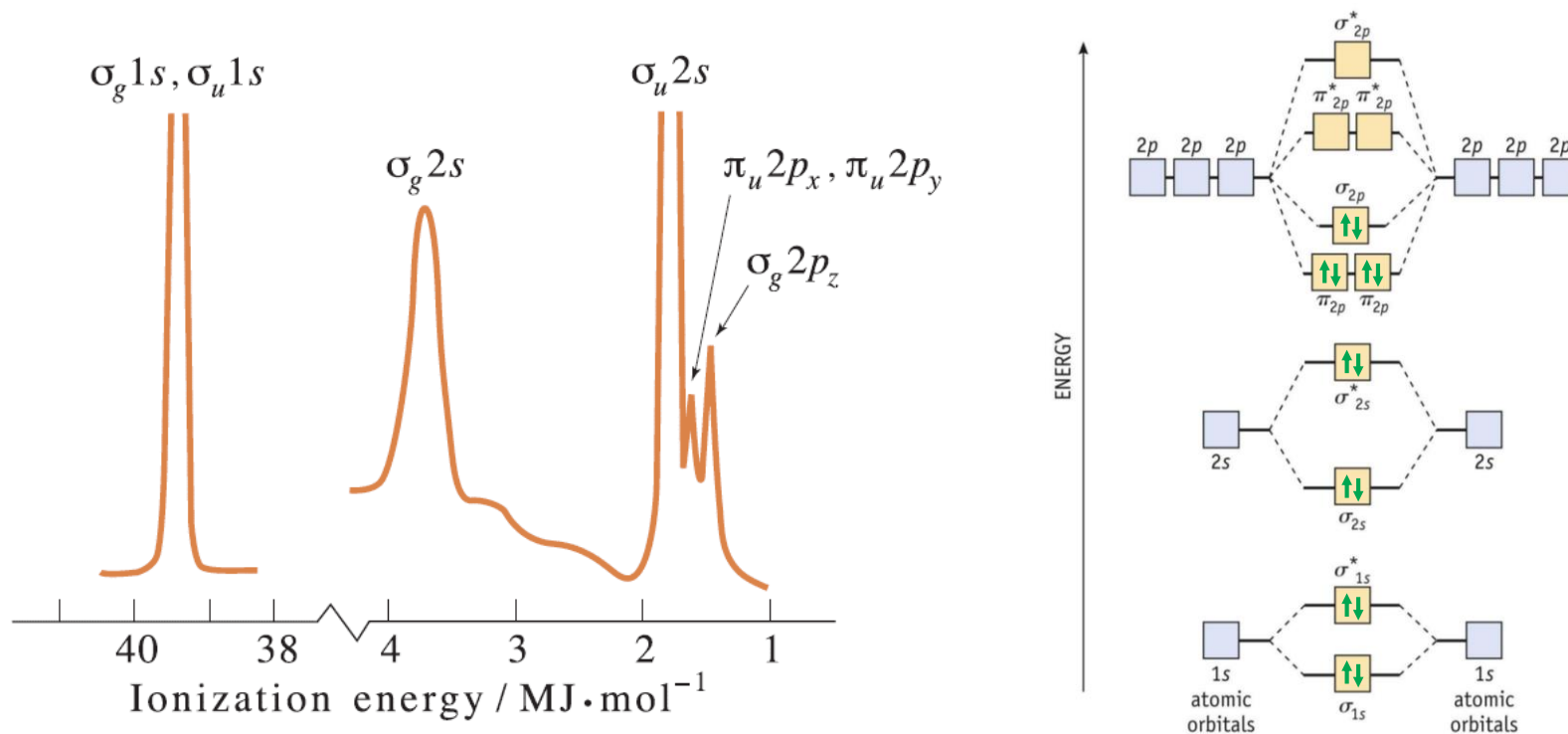
M 11.2

Homonuclear Diatomic Molecules



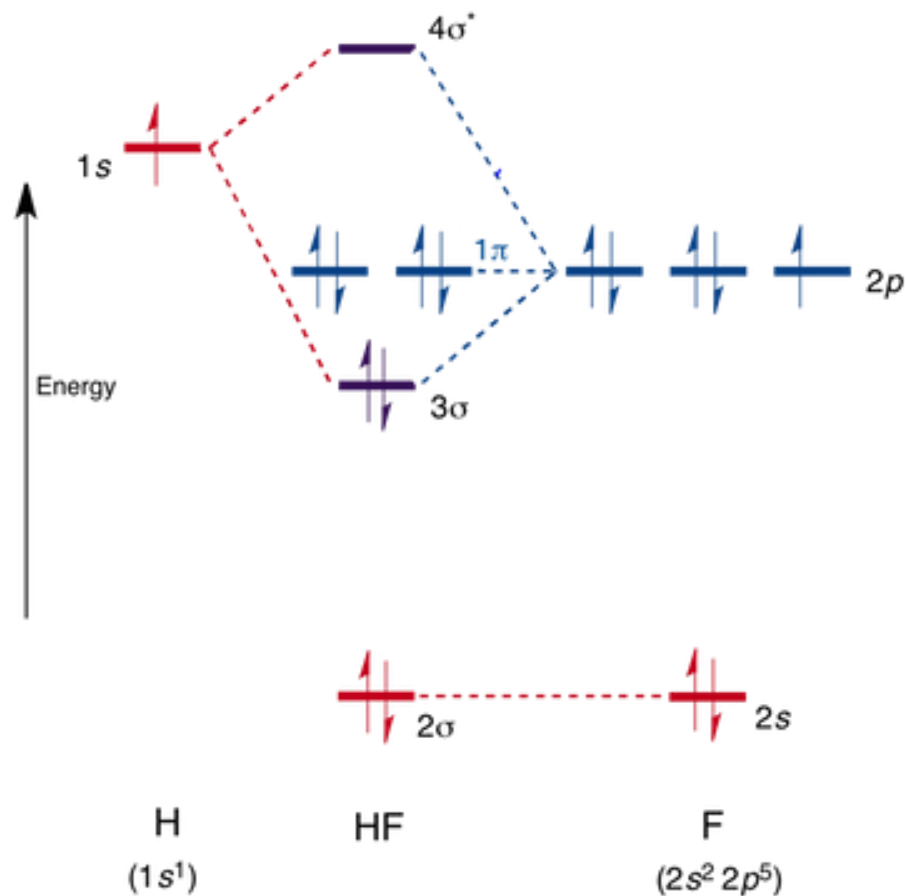
M 11.2

Photoelectron Spectrum of N₂



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.7)

Heteronuclear Case: HF



(Image: <https://www.chemtube3d.com/orbitalshf/>)